

NSEE INDIA — MASTER PROJECT BRIEF

Complete Technical Handoff Document & Feature Specifications for Development Team

VERSION & DATE

v1.0 • October 2026

DOMAIN & PORTAL

nseeindia.com

TARGET AUDIENCE

Development Team / Agents

STATUS

Ready for Sprint Execution

1. PROJECT OVERVIEW & SCOPE

NSEE India is a high-availability, government-grade online examination and scholarship discovery portal engineered to conduct entrance and talent exams across all Indian states and union territories, managed district-wise, supporting both **online proctored** and **offline centre-based** delivery modes.

STUDENT LIFECYCLE: Register & OTP → Auto USID → Progressive Profile → Apply → Pay (₹99) → Admit Card (QR) → Exam → Result & Rank

Exam Levels Covered

- **School Levels:** Class 5, 6, 7, 8, 9, 10, 11, 12
- **Higher Education:** BCA, B.Tech, MCA, M.Tech
- **Exam Fee:** ₹99 flat per exam (Razorpay gateway)

Launch Strategy & Timelines

- **Phase 1 Public Launch (Week 7):** Core Auth, Registration, USID, Profile, Public Notice Board, Static Catalog.
- **Full Platform (Month 9):** Question bank, auto paper generation, live AI proctoring, offline centre sync & analytics.

2. FINAL TECHNOLOGY STACK

LAYER / DOMAIN	SELECTED TECHNOLOGY & VERSION	ARCHITECTURAL SCOPE / NOTES
Backend Core	Java 21 LTS + Spring Boot 3.3 (Maven)	Hexagonal (Ports & Adapters) architecture
Persistence & DB	PostgreSQL 16 + Spring Data JPA + Hibernate + Flyway	Strict Flyway migrations, reversible scripts
Caching & Sessions	Redis 7	OTP lifecycle, session tokens, rate limiting
Security & Auth	Spring Security + Stateless JWT + SMS/Email OTP	BCrypt hashing, 15m access token, 7d refresh token
Payments	Razorpay Java SDK	Server-side order generation & webhook verification
PDF Engine	OpenPDF	Dynamic Admit Cards with QR code verification
Real-Time Comms	Spring WebSocket (STOMP protocol)	Active proctor dashboard & live session timer sync

LAYER / DOMAIN	SELECTED TECHNOLOGY & VERSION	ARCHITECTURAL SCOPE / NOTES
Frontend Web	React 18 + TypeScript + Vite + Tailwind CSS + TanStack Query	Responsive, government-modern UI with RTL/i18n
Infra & DevOps	Docker Compose + Nginx + GitHub Actions + VPS	Strictly no Kubernetes / Kafka initially for operational simplicity
Object Storage	S3-compatible (MinIO / AWS S3)	Encrypted candidate identity & document verification vaults

3. ARCHITECTURE: HEXAGONAL (PORTS & ADAPTERS)

The system enforces strict boundary isolation. Business rules reside exclusively inside the domain core with zero dependency on frameworks or persistence layers.

Layer Boundaries

- `domain/` : Pure business models, aggregates, domain events, and ports. Imports JDK only.
- `application/` : Use case implementations, orchestrators, command/query DTOs.
- `infrastructure/` : Implementations of outbound ports (JPA repositories, Redis adapters, Razorpay, S3, SMS clients).
- `presentation/` : Spring REST Controllers invoking inbound ports only.

Feature-First Package Layout

```
src/main/java/com/nsee/
├─ core/                # Shared kernel & cross-cutting
├─ features/
│   ├─ auth/            # Login, OTP, JWT, RBAC
│   └─ student/         # Registration & USID
sequence
│   ├─ profile/         # 6-step progressive profile
│   ├─ exam/            # State machine lifecycle
│   ├─ questionbank/    # Questions & bulk CSV
│   ├─ paper/           # Randomized sets (A/B/C/D)
│   ├─ application/     # Application & eligibility
│   ├─ payment/         # Razorpay integration
│   ├─ admitcard/       # OpenPDF QR generator
│   ├─ examsession/     # Real-time exam runner
│   ├─ proctoring/      # Anti-cheating violations
│   ├─ result/          # Rank calculations & scores
│   └─ notice/          # Targeted public bulletins
└─ bootstrap/          # Spring Boot main
application
```

4. ROLES & PERMISSIONS HIERARCHY

ROLE IDENTIFIER	ADMINISTRATIVE SCOPE	CORE RESPONSIBILITIES & CAPABILITIES
<code>SUPER_ADMIN</code>	Global / National	Create admins, final approve exams/notices (Maker-Checker), global configuration, full audit inspection.
<code>EXAM_SETTER</code>	State / National	Curate question banks, draft exams, configure sets, schedule windows. (Cannot approve own exams).
<code>STATE_ADMIN</code>	State Level	Oversee district controllers, monitor state quotas, inspect district compliance and aggregated reports.
<code>DISTRICT_CONTROLLER</code>	District Level	Onboard and verify physical exam centres, assign Centre Controllers, allocate candidate cohorts.

ROLE IDENTIFIER	ADMINISTRATIVE SCOPE	CORE RESPONSIBILITIES & CAPABILITIES
CENTRE_CONTROLLER	Physical Centre	Verify candidate biometric/photo identity on-site, assign test terminals, mark physical attendance, unlock exam sessions.
INVIGILATOR / PROCTOR	Centre / Remote	Live dashboard monitoring, broadcast warnings, register violations, trigger emergency exam termination.
STUDENT	Self	Self-registration, profile completion, exam application, payment, admit card retrieval, exam taking, score inspection.
SUPPORT / AUDITOR	Read-Only Audit	Manage public grievance tickets, resolve candidate inquiries, inspect immutable audit and security logs.

5. PUBLIC-FIRST LAUNCH STRATEGY (WEEK 7 TARGET)

CRITICAL MANDATE: Launch Sprint 0 through Sprint 3 by Week 7. The general public must experience a production-ready, fully responsive portal establishing trust, candidate registration, and scholarship visibility.

Public Portal Deliverables (No Login Required):

- Government-style accessible homepage with National Emblem, bilingual toggle (English/Hindi), and font resizer.
- Official Notice Board featuring category filtering (General, Circular, Alert, Exam Schedules) with PDF downloads.
- Exam Catalog displaying upcoming levels with a prominent "Notify Me" trigger (applications gated by feature flag).
- Public Query & Grievance form with ticket tracking ID and email acknowledgement.
- Mandatory Compliance Pages: RTI, Terms of Service, Privacy Policy, Grievance Officer details, and Sitemap.

Feature Flag Configuration (application.yml)

```
features:
  auth: true           # Active for student onboarding
  student: true        # Active for registration & USID issuance
  profile: true        # Active for 6-stage verification
  notice: true         # Active for public advisories
  query: true          # Active for public ticketing
  exam: false          # Catalog displayed via read-only static JSON
  application: false   # Gated until Phase 2
  payment: false       # Gated until Phase 2
  examsession: false   # Gated until Phase 3
  proctoring: false    # Gated until Phase 3
  result: false        # Gated until Phase 4
```

6. DETAILED 20-SPRINT IMPLEMENTATION ROADMAP

SPRINT	FEATURE MODULE	DURATION	DELIVERABLES & ACCEPTANCE CRITERIA
Sprint 0	Core Foundation	1.0 Wk	Hexagonal scaffold, Flyway migrations, Docker Compose (Postgres, Redis, Nginx), CI pipeline with Testcontainers, <code>/actuator/health</code> .
Sprint 1	Auth & RBAC	2.0 Wk	Spring Security + JWT, SMS/Email OTP generation and verification, Redis session store, login audit logging, role protection.

SPRINT	FEATURE MODULE	DURATION	DELIVERABLES & ACCEPTANCE CRITERIA
Sprint 2	Student & USID	2.0 Wk	Student registration with auto-generated USID (NSEE/YYYY/STATE/DIST/000123) via atomic sequence generator. Partial save support.
Sprint 3	Progressive Profile	2.0 Wk	6-stage profile: Personal, Academic, Parental, Bank, Address, Documents. Autosave, completeness percentage calculation.
 MILESTONE: PUBLIC LAUNCH AT WEEK 7 (Public Portal, Registrations, USID, Profile Engine Live)			
Sprint 4	Geography & Centres	1.5 Wk	State, District, and Exam Centre hierarchies; Centre Controller onboarding, capacity mapping, and physical room setup.
Sprint 5	Notice Board	1.0 Wk	Notice creation workflow (Draft → Review → Approved → Published). Multi-level geographic targeting, document attachments, home widget.
Sprint 6	Public Query Form	1.0 Wk	Public ticketing system without authentication requirement, automated ticket token generation, CAPTCHA, admin response hub.
Sprint 7	Exam Lifecycle CRUD	2.0 Wk	Maker-Checker state machine: DRAFT → PENDING → APPROVED → PUBLISHED . Setter cannot approve own exam. Fee & schedule gating.
Sprint 8	Question Bank	2.0 Wk	MCQ storage, tag taxonomy (Subject, Topic, Difficulty, Language), bulk CSV/Excel parsing with rollback and schema validation.
Sprint 9	Paper Generation	2.0 Wk	Automated set generator (Sets A, B, C, D). Question and option permutation algorithm, AES-256 payload encryption prior to exam schedule.
Sprint 10	Exam Application	1.5 Wk	Eligibility validator checking academic stage against exam requirements; preference selector for 3 exam centres, draft application state.
Sprint 11	Payment (Razorpay)	2.0 Wk	Server-side Razorpay order API, webhook listener with HMAC signature verification, idempotent transaction updates, PDF receipt creation.
Sprint 12	Admit Card Generator	1.5 Wk	OpenPDF generator creating encrypted admit card PDFs with signed QR verification codes, seat allocations, reporting instructions, release gating.
Sprint 13	Exam Session Engine	3.0 Wk	WebSocket session synchronization, 5-second delta answer autosave, persistent server countdown timer, auto-submission on timeout. 1,000+ load tested.
Sprint 14	Anti-Cheating & Proctoring	2.0 Wk	Browser lock: tab-switch, copy-paste, fullscreen detector. WebRTC snapshots for multi-face/no-face checks. 3-strike siren auto-submit. Proctor room.
Sprint 15	Evaluation & Results	1.5 Wk	Deterministic MCQ scoring, percentile normalization across paper sets, district/state/national rank generation, public scorecards.
Sprint 16	Notification Engine	1.5 Wk	Event-driven communications via Email, SMS, WhatsApp. Templated dispatch, retry queue with Dead Letter Queue (DLQ) support.
Sprint 17	Admin Central Hub	2.0 Wk	Super Admin operational console, role escalation, approval workflows, system parameter overrides, real-time platform metrics.

SPRINT	FEATURE MODULE	DURATION	DELIVERABLES & ACCEPTANCE CRITERIA
Sprint 18	Compliance & Audit	1.0 Wk	Immutable append-only audit trail via <code>@Auditable</code> aspect, logging IP, actor, entity ID, previous payload, and diff across every mutation.
Sprint 19	Offline Centre App	2.0 Wk	Offline centre verification portal: batch candidate check-in, biometric/photo matching, physical terminal unlock, local attendance sync.
Sprint 20	Analytics & Reports	1.5 Wk	PostgreSQL Materialized Views for state/district score distributions, candidate participation rates, proctor violation audits, CSV/PDF reports.

7. CORE TECHNICAL SPECIFICATIONS

Universal Scholarship ID (USID)

Every registered student receives an immutable, globally unique USID structured as:

```
NSEE / {YYYY} / {STATE_CODE} / {DIST_CODE} / {SEQ_NUM:06d}
Example: NSEE/2026/UP/GNO/000142
```

Implemented using a pessimistic locking sequence table to prevent race conditions during peak registration concurrency.

Stateless Security & Rate Limiting

- **Access Tokens:** 15-minute expiration, signed with RSA-256 or high-entropy HMAC-SHA512.
- **Refresh Tokens:** 7-day expiration, persisted in Redis with device fingerprinting and automatic rotation on reuse.
- **Rate Limiting:** Redis token-bucket via Bucket4j restricting public auth/OTP endpoints to 5 calls/min per IP.

Proctoring & Anti-Cheating Protocol

Violation Threshold: Maximum 3 strikes before immediate automated exam termination.

- **Screen Focus:** Page Visibility API detecting blur and tab switches.
- **Fullscreen Enforcement:** Exiting full screen triggers an audible siren and blocking modal.
- **Clipboard Isolation:** Complete suppression of context menus, copy, paste, and text selection events.
- **Visual AI:** Periodic webcam sampling to flag presence of multiple faces, no face, or mobile devices.
- **Live Proctor Hub:** STOMP over WebSocket channels pushing incident events to active invigilator consoles.

8. DATABASE ARCHITECTURE & KEY RELATIONAL SCHEMAS

All database modifications are strictly version-controlled via Flyway scripts (`v1__init.sql` , `v2__auth.sql`). Auto-DDL updates in production are strictly forbidden.

```

-- Core Authentication & Access Control
users (id, email, phone, password_hash, status, created_at, updated_at)
roles (id, code, name, description)
user_roles (user_id, role_id)
otp_verifications (id, identifier, otp_code, type, expires_at, verified)
refresh_tokens (id, user_id, token_hash, device_info, expires_at)

-- Candidates, Identity & Profiles
students (id, user_id, usid, state_id, district_id, registration_year, current_step)
student_profiles (id, student_id, personal_info, academic_info, bank_info, address_info,
completeness_pct)
profile_documents (id, profile_id, doc_type, s3_key, file_size, verified_status)

-- Administrative Geography & Physical Centres
states (id, code, name)
districts (id, state_id, code, name)
centres (id, district_id, name, code, address, capacity_per_slot, status)
centre_attendance (id, centre_id, student_id, exam_id, terminal_no, check_in_time, status)

-- Examination Management & Question Banking
exams (id, code, title, level, fee_amount, registration_start, registration_end, exam_date, status)
questions (id, subject_id, class_level, question_type, body_html, marks, penalty_marks, language)
question_options (id, question_id, option_key, body_html, is_correct)
exam_papers (id, exam_id, set_code, encrypted_payload_key, total_questions, duration_minutes)

-- Applications, Transactions & Live Execution
exam_applications (id, student_id, exam_id, status, payment_id, centre_pref_id)
payments (id, order_id, payment_id, amount, currency, status, webhook_payload, receipt_url)
admit_cards (id, application_id, qr_signature, verification_url, seat_number, is_released)
exam_sessions (id, student_id, paper_id, start_time, end_time, remaining_seconds, is_submitted)
session_responses (id, session_id, question_id, selected_option_id, marked_for_review, last_updated)
proctoring_violations (id, session_id, violation_type, snapshot_s3_key, timestamp, proctor_flag)
results (id, exam_id, student_id, raw_score, normalized_score, district_rank, state_rank,
national_rank)
audit_logs (id, actor_id, actor_role, action, entity_name, entity_id, previous_state, new_state,
ip_address, created_at)

```

9. FRONTEND ARCHITECTURE & DESIGN GUIDELINES

Brand Color Palette (Gov-Modern)

TOKEN NAME	HEX VALUE	PRIMARY USAGE
Navy Blue	#0B3D91	Header, primary buttons, branding
India Saffron	#FF9933	Accents, banners, primary highlights
India Green	#138808	Success badges, verified states
Page BG	#F7F9FC	Light, glare-free application background
Surface / Card	#FFFFFF	Content containers, elevated cards
Text Dark	#1A1A1A	High contrast readable body copy

Typography & Accessibility Standards

- **Headings:** Noto Serif / Merriweather for institutional gravitas.
- **Body UI:** Noto Sans / Inter for clear legibility on mobile screens.
- **Indic Scripts:** Noto Sans Devanagari for Hindi and regional scripts.
- **Compliance:** Strict conformance with **GIGW** (Guidelines for Indian Government Websites) and **WCAG 2.1 AA**.
- **Features:** Minimum 4.5:1 color contrast ratio, full keyboard tab navigation, ARIA live regions, skip-to-content links.

10. API CONVENTIONS & STANDARD ERROR CONTRACT

All endpoints are prefixed under `/api/v1`. Every request and response conforms to UTF-8 JSON. Standardized error payloads must be returned on all HTTP 4xx and 5xx conditions:

```
{
  "timestamp": "2026-10-04T13:22:00Z",
  "status": 400,
  "error": "VALIDATION_FAILED",
  "message": "Email address already registered under another USID",
  "path": "/api/v1/students/register",
  "details": [
    { "field": "email", "issue": "UniqueConstraintViolation" }
  ]
}
```

11. CI/CD PIPELINE & VPS DEPLOYMENT

Continuous Integration is executed on GitHub Actions across every push and Pull Request. Pull requests cannot be merged unless all test gates pass with minimum 85% code coverage.

CI Pipeline Validation Steps

- 1. Format check: `mvn spotless:check`
- 2. Java compilation: `mvn compile`
- 3. Unit testing: `mvn test` (Fails if line coverage < 85%)
- 4. Integration testing: `mvn verify` with real PostgreSQL & Redis via Testcontainers
- 5. E2E API test suites: REST Assured execution
- 6. Frontend lint & test: ESLint + Vitest + React Testing Lib
- 7. Frontend production build: `npm run build`

Staging / Production Deployment Flow

- 1. Docker multi-stage build creates distroless runtime container.
- 2. Container pushed to GitHub Container Registry (ghcr.io).
- 3. SSH trigger initiates pull on production VPS.
- 4. Docker Compose restarts containers with zero downtime.
- 5. Flyway executes pending database migrations on startup.
- 6. Spring Actuator health check verifies operational readiness.

12. DEFINITION OF DONE (DOD) PER FEATURE

A feature is considered completed and eligible for main branch merging only when all criteria below are verified:

- ☐ OpenAPI 3.0 / Swagger specification defined, reviewed, and merged into docs.
- ☐ Flyway database migrations committed, tested for roll-forward and rollback integrity.
- ☐ Domain logic unit test coverage verified at ≥ 90% in isolation.
- ☐ Integration tests passing with Testcontainers (PostgreSQL 16 + Redis 7).
- ☐ React UI component responsive across mobile, tablet, and desktop viewports.
- ☐ Accessibility checks validated (WCAG 2.1 AA keyboard navigation and screen reader labels).
- ☐ Audit logging registered and verified for every state mutation.
- ☐ Feature flag entry created and set to disabled by default.
- ☐ Deployed and smoke-tested on staging environment with stakeholder sign-off.

13. SUCCESS METRICS & PERFORMANCE KPIS

OPERATIONAL METRIC	TARGET THRESHOLD	MONITORING METHOD
Public Platform Launch	Week 7 (Strict Milestone)	Production release tag deployment
Candidate Registration Flow	< 2 Minutes Total Duration	Client session conversion analytics
Profile Completion Rate	> 80% of registered students	Cohort funnel analytics
Backend API p95 Latency	< 300 Milliseconds	Prometheus / Grafana metrics
Exam Session Concurrency	10,000+ Simultaneous Candidates	JMeter / Gatling distributed load test
Live Exam System Uptime	99.9% Availability SLA	Uptime monitoring & health check probe
Accessibility Compliance	WCAG 2.1 AA Certified	Automated Axe / Lighthouse auditing
Initial Page Load Time	< 2.0 Seconds over 4G	Web Vitals LCP tracking

Next Immediate Action: Initialize Sprint 0 scaffold repository immediately using Spring Boot 3.3 and Java 21 LTS. Construct hexagonal root packages and configure Docker Compose with PostgreSQL 16 and Redis 7. All technical questions and architectural variations must align with this single source of truth.